

$p = (p_1, p_2, \dots, p_n)$ pmf of phrases (malicious)

$T = \{t_1, t_2, \dots, t_n\}$ set of phrases

$S = \{s_1, s_2, \dots, s_n\}$ set of $p(t_i \text{ works} | \text{unflagged})$

$q = (q_1, \dots, q_n)$ pmf of phrases (natural)

Flag when $D_{KL}(p || q) = \sum_{i=1}^n p_i \log \left(\frac{p_i}{q_i} \right) > C$

Want p to maximize $\sum_{i=1}^n p_i s_i$ w/ constraint

$D_{KL}(p || q) \leq C$ (assume q , C known)
this may be unreasonable

$$\mathcal{L}(p, \lambda, \mu) = \sum_i p_i s_i - \lambda \left(\sum_i p_i - 1 \right) - \mu \left(\sum_i p_i \log \left(\frac{p_i}{q_i} \right) - C \right)$$

$$\frac{\partial \mathcal{L}}{\partial p_i} = s_i - \lambda - \mu \left(\log \left(\frac{p_i}{q_i} \right) + 1 \right) = 0$$

$$\log \left(\frac{p_i}{q_i} \right) = \frac{s_i - \lambda}{\mu} - 1$$

$$\frac{p_i}{q_i} = e^{\frac{s_i - \lambda}{\mu} - 1} \Rightarrow p_i = q_i e^{\frac{s_i - \lambda}{\mu} - 1}$$

$$\sum_i p_i = 1 \Rightarrow \sum_i q_i e^{\frac{s_i - \lambda}{\mu} - 1} = 1$$

$$\sum_i q_i e^{\frac{s_i}{\mu}} e^{-\frac{\lambda}{\mu} - 1} = 1$$

$$e^{-\frac{\lambda}{\mu} - 1} = \frac{1}{\sum_i q_i e^{\frac{s_i}{\mu}}} \Rightarrow -\frac{\lambda}{\mu} = 1 - \log \left(\sum_i q_i e^{\frac{s_i}{\mu}} \right)$$

$$\Rightarrow \lambda = \mu \log \left(\sum_i q_i e^{\frac{s_i}{\mu}} \right) - \mu$$

$$P_i = q_i e^{\frac{s_i - \lambda}{\mu}} - 1$$

$$\lambda = \mu \log \left(\sum_j q_j e^{\frac{s_j}{\mu}} \right) - \mu$$

$$P_i = q_i e^{\frac{s_i - \mu \log \left(\sum_j q_j e^{\frac{s_j}{\mu}} \right) + \mu}{\mu}} - 1$$

$$= q_i e^{\frac{s_i}{\mu}} e^{-\log \left(\sum_j q_j e^{\frac{s_j}{\mu}} \right)}$$

$$= \frac{q_i e^{\frac{s_i}{\mu}}}{\sum_j q_j e^{\frac{s_j}{\mu}}}$$

$$P_i = \frac{q_i e^{\frac{s_i}{\mu}}}{X(\mu)}$$

$$\hookrightarrow X(\mu)$$

$$= \sum_j q_j e^{\frac{s_j}{\mu}}$$

$$\sum_i P_i \log \left(\frac{P_i}{q_i} \right) = C$$

$$\sum_i \frac{q_i e^{\frac{s_i}{\mu}}}{\sum_j q_j e^{\frac{s_j}{\mu}}}$$



$$\log \left(\frac{q_i e^{\frac{s_i}{\mu}}}{\sum_j q_j e^{\frac{s_j}{\mu}}} \right) = C$$

$$= \frac{s_i}{\mu} + \log(q_i) - \log \left(\sum_j q_j e^{\frac{s_j}{\mu}} \right) - \log(q_i)$$

$$\sum_i \frac{q_i s_i e^{\frac{s_i}{\mu}}}{\mu \sum_j q_j e^{\frac{s_j}{\mu}}} - \frac{q_i e^{\frac{s_i}{\mu}} \log \left(\sum_j q_j e^{\frac{s_j}{\mu}} \right)}{\sum_j q_j e^{\frac{s_j}{\mu}}} = C$$

$$= \frac{1}{\mu} \sum_i P_i s_i - P_i \log \left(\sum_j q_j e^{\frac{s_j}{\mu}} \right) = C$$

$$\Rightarrow \mu = \frac{\sum_i P_i (s_i - \log(X))}{C}$$